

Vadivelan Murugesan

vadivelamurugesan@gmail.com | 347 222 8351 | Portfolio | LinkedIn | GitHub

Skills

Programming:	Python, C, C++, Lua
Robotics & Control:	ROS 2, MoveIt, Control Systems, Kinematics, Teleoperation
Embedded Systems:	STM32, Arduino, Raspberry Pi, Jetson Orin Nano, ODrive, Serial Protocols (CAN, SPI, I2C), PCB Debugging
Perception & CV:	OpenCV, ArUco, YOLOv8, ICP, RANSAC
ML & Frameworks:	PyTorch, TensorFlow, Scikit-learn, CUDA, Autoencoder
Simulation Tools:	Gazebo, CoppeliaSim, MATLAB, Simulink
Sensors & Actuators:	RGB and Depth Cameras, IMU, LiDAR, Actuators (Servo, BLDC)

Education

New York University , MS in Mechatronics and Robotics	Sep 2023 - May 2025
Anna University, Chennai , BE in Electronics and Communication	Sep 2018 - June 2022

Experience

Research Scholar , NYU Mechanical & Aerospace Engineering – New York, NY	Mar 2026 – Present
- Integrating Autoware Universe on a Jetson AGX Orin with an Ouster OS0-32 LiDAR, resolving TF tree misalignments, custom pointcloud format conversion (PointXYZIRC), and NDT scan matching pipeline issues for real-world localization. - Built a GNSS-free LiO-SAM map of the NYU area and configured Autoware's localization stack for local projection, enabling NDT-based pose estimation without GPS infrastructure. - Developed a ROS 2 converter node to reformat Ouster pointclouds into Autoware-compatible PointXYZIRC format, bridging driver output with the crop box filter and downstream perception pipeline. - Configured CycloneDDS unicast networking for cross-machine ROS 2 communication between Jetson AGX Orin and an x86 workstation, enabling distributed sensor processing and stack validation.	
System Engineer Intern , Gen Auto AI – Piscataway, NJ	Nov 2025 – Mar 2026
- Built production-quality localization infrastructure from scratch: collected Ouster LiDAR data via rosbag, generated PCD maps with LIO-SAM, and authored Lanelet2 HD maps, cutting environment setup time by ~60%. - Closed the full autonomy loop on an electric go-kart by integrating perception, localization, planning, and control under Autoware, enabling the first successful real-world autonomous navigation runs. - Reduced data re-collection cycles by ~40% by standardizing sensor data pipelines across environments, establishing a consistent performance evaluation baseline.	
Research Scientist , ASAS Lab, NYU – New York, NY	Sep 2025 – Nov 2025
- Reduced time-to-first-test from days to under an hour by authoring reusable Autoware + AWSIM scenario workflows in ROS 2 on Jetson AGX Orin, adopted by 3 other researchers in the lab. - Doubled experiment throughput by building modular test harnesses for scenario testing, sensor validation, and algorithm benchmarking that could be reconfigured without modifying core stack code. - Prevented hardware-damaging bugs from reaching physical vehicles by monitoring simulation runs and resolving perception, localization, and navigation failures before any physical deployment.	
Robotics Engineer , Surinova Pvt Ltd – Chennai, India	Nov 2022 – Aug 2023
- Achieved real-time synchronized multi-motor control by integrating STM32F4 with ODrive over CAN, SPI, and I2C, eliminating actuation lag that caused assembly failures in the prior hardware revision. - Caught 5+ critical firmware bugs before field deployment by building a hardware-in-the-loop testbed with multiple ODrives and an STM32, preventing at least 2 weeks of costly on-site debugging. - Cut troubleshooting time by 50% by deploying PicoScope-based signal analysis and custom debug scripts to isolate encoder and SPI noise, a recurring failure mode that had stalled the team for months.	

Technical Intern, Cologenes Healthcare Pvt Ltd – Salem, India

July 2021 – Sep 2021

- Reduced production processing time by 30% and increased throughput by 20% by automating a collagen sheet production step with a laser-based workflow, removing a manual bottleneck.
- Improved component alignment accuracy by 80% by implementing a computer vision inspection and positioning system, replacing error-prone manual placement.

Projects

Proactive Human Manipulator Collision Avoidance

GitHub

- Built a digital twin in Gazebo with real time 3D human pose estimation using an RGB D camera to simulate and test safe manipulator behavior in dynamic, human populated settings.
- Integrated ROS 2, MoveIt, and a UR16e robotic arm for adaptive, human aware motion control.
- Implemented an A RRT* planner for dynamic path planning, enabling safe trajectory replanning based on predicted human motion.

Automated Object Retrieval Mobile Robot

GitHub

- Developed a mobile robot with camera and Raspberry Pi for real time object identification using OpenCV; designed detection and localization.
- Engineered and programmed a two DOF arm to autonomously navigate and relocate objects within an arena, demonstrating control and planning in dynamic environments.

Color Cube Sorting Robotic Arm

GitHub

- Built a four DOF robotic arm with a gripper to sort color cubes on a conveyor; integrating a color sensor for real time identification and an ultrasonic sensor for proximity.
- Improved efficiency by utilizing cogs on the Parallax Propeller Board for multiprocessing, enabling simultaneous color and distance sensing.

Teleoperation of a 4 DOF Robotic Arm

GitHub

- Designed and prototyped a 4 DOF robotic arm controllable via a puppet rig with four potentiometers, similar to industrial pick and place.
- Added program memory for up to ten reprogrammable positions with simple save and cycle for flexible operation.
- Built a real time mirror mode that replicates the puppet arm's orientation for live teleoperation.

Key Projects in Perception, Machine Learning, and 3D Mapping

GitHub

- Developed custom RANSAC plane fitting and Iterative Closest Point for 3D point clouds to achieve accurate plane extraction and point cloud registration.
- Built augmented reality visualizations in OpenCV with ArUco markers, projecting 3D objects onto real world tags with calibrated rendering.
- Implemented object detection and tracking with YOLOv8 and a tracking pipeline to maintain persistent IDs across frames in dynamic video.

Robot Perception: Solving a 3D Maze Using Graph Networks

- Built a Spatial Vicinity Graph that treats each point in the 3D maze as a node with edges formed from global image descriptors, enabling shortest path planning with 100 percent accuracy in tests.
- Used SuperPoint and SuperGlue for feature extraction and matching with geometric verification to identify the destination node for precise navigation.
- Evaluated COLMAP, pose graph optimization with GTSAM, and Stella VSLAM before converging on a vision plus graph theory approach that balanced accuracy and runtime.